

2.1 Integration Worksheet

1. Find
- $\int x e^{-3x} dx$
- .

Let $u = x$ and $dv = e^{-3x} dx$

These two equations imply $du = dx$ & $v = -\frac{1}{3}e^{-3x}$.

Therefore, by IBP we get

$$\int x e^{-3x} dx = -\frac{1}{3} x e^{-3x} - \int -\frac{1}{3} e^{-3x} dx = -\frac{1}{3} x e^{-3x} + \int \frac{1}{3} e^{-3x} dx = -\frac{1}{3} x e^{-3x} - \frac{1}{9} e^{-3x} + C$$

2. Find
- $\int \frac{\ln(x)}{x^2} dx$

We let $u = \ln(x)$ and $dv = \frac{1}{x^2} dx$.

These two equations imply: $du = \frac{1}{x} dx$ and $v = -\frac{1}{x}$.

Now, we apply the integration by parts formula and we obtain:

$$\int \frac{\ln(x)}{x^2} dx = -\frac{1}{x} \ln(x) - \int -\frac{1}{x} \left(\frac{1}{x} \right) dx = -\frac{1}{x} \ln(x) + \int \frac{1}{x^2} dx = -\frac{1}{x} \ln(x) - \frac{1}{x} + C$$

3. Find
- $\int x^2 \cos(x) dx$
- .

Let $u = x^2$ and $dv = \cos(x) dx$

These two equations imply $du = 2x dx$ & $v = \sin(x)$.

Therefore, by IBP we get

$$\int x^2 \cos(x) dx = x^2 \sin(x) - \int \sin(x) 2x dx$$

On the new integral we let $u = 2x$ and $dv = \sin(x) dx$

These two equations imply $du = 2 dx$ & $v = -\cos(x)$

$$\begin{aligned} x^2 \sin(x) - \int \sin(x) 2x dx &= x^2 \sin(x) - \left[-2x \cos(x) - \int -2 \cos(x) dx \right] \\ &= x^2 \sin(x) + 2x \cos(x) + \int 2 \cos(x) dx \\ &= x^2 \sin(x) + 2x \cos(x) + 2 \sin(x) + C \end{aligned}$$

4. $\int \ln(x^2 + 1) dx$

Let $u = \ln(x^2 + 1)$ and $dv = dx$.

These two equations imply $du = \frac{2x}{x^2 + 1} dx$ & $v = x$.

Therefore, by IBP, we get

$$\begin{aligned} \int \ln(x^2 + 1) dx &= x \ln(x^2 + 1) - \int \frac{2x^2}{x^2 + 1} dx = x \ln(x^2 + 1) - \int \frac{2x^2 + (2 - 2)}{x^2 + 1} dx \\ &= x \ln(x^2 + 1) - \int \left(\frac{2x^2 + 2}{x^2 + 1} + \frac{-2}{x^2 + 1} \right) dx = x \ln(x^2 + 1) - \int \left(2 + \frac{-2}{x^2 + 1} \right) dx \\ &= x \ln(x^2 + 1) - (2x - \tan^{-1}(x)) + C \\ &= x \ln(x^2 + 1) - 2x + \tan^{-1}(x) + C \end{aligned}$$

5. $\int (x^2 + x + 1) \sin(x) dx$

We can use Tabular IBP here since one of the factors in our integral is a polynomial and the other we can easily find antiderivatives for.

$P(x)$	$F(x)$	Sign +/-
$x^2 + x + 1$	$\sin(x)$	Skip
$2x + 1$	$-\cos(x)$	+
2	$-\sin(x)$	-
0	$\cos(x)$	+

Therefore, by Tabular IBP, we get

$$\int (x^2 + x + 1) \sin(x) dx = -(x^2 + x + 1) \cos(x) + (2x + 1) \sin(x) + 2 \cos(x) + C$$

6. Find $\int_{\sqrt{e}}^e \frac{\ln(x)}{x^2} dx$.

Let $u = \ln(x)$ and $dv = \frac{1}{x^2} dx$.

These two equations imply $du = \frac{1}{x} dx$ & $v = -\frac{1}{x}$.

Therefore, by IBP, we get

$$\begin{aligned} \int_{\sqrt{e}}^e \frac{\ln(x)}{x^2} dx &= -\frac{\ln(x)}{x} \Big|_{\sqrt{e}}^e - \int_{\sqrt{e}}^e -\frac{1}{x^2} dx = \left(-\frac{\ln(e)}{e} + \frac{\ln(\sqrt{e})}{\sqrt{e}} \right) + \int_{\sqrt{e}}^e \frac{1}{x^2} dx \\ &= \left(-\frac{1}{e} + \frac{\frac{1}{2}}{\sqrt{e}} \right) - \frac{1}{x} \Big|_{\sqrt{e}}^e = \left(-\frac{1}{e} + \frac{\frac{1}{2}}{\sqrt{e}} \right) - \left(\frac{1}{e} - \frac{1}{\sqrt{e}} \right) = -\frac{2}{e} + \frac{3}{2\sqrt{e}} \end{aligned}$$

7. $\int x^3 \sqrt{2x+1} dx$

We can use Tabular IBP here since one of the factors in our integral is a polynomial and the other we can easily find antiderivatives for.

$P(x)$	$F(x)$	Sign +/-
x^3	$\sqrt{2x+1}$	Skip
$3x^2$	$\frac{1}{3}(2x+1)^{3/2}$	+
$6x$	$\frac{1}{3 \cdot 5}(2x+1)^{5/2}$	-
6	$\frac{1}{3 \cdot 5 \cdot 7}(2x+1)^{7/2}$	+
0	$\frac{1}{3 \cdot 5 \cdot 7 \cdot 9}(2x+1)^{9/2}$	-

Therefore, by Tabular IBP, we get

$$\int x^3 \sqrt{2x+1} dx = \frac{x^3}{3} (2x+1)^{3/2} - \frac{x^2}{5} (2x+1)^{5/2} + \frac{2x}{5 \cdot 7} (2x+1)^{7/2} - \frac{2}{5 \cdot 7 \cdot 9} (2x+1)^{9/2} + C$$

8. Calculate the exact value of: $\int_0^1 \tan^{-1}(x) dx$

We let $u = \tan^{-1}(x)$ and $dv = dx$.

These two equations imply: $du = \frac{1}{x^2 + 1} dx$ and $v = x$.

Now, we apply the integration by parts formula and we obtain:

$$\int_0^1 \tan^{-1}(x) dx = x \tan^{-1}(x) \Big|_0^1 - \int_0^1 \frac{x}{x^2 + 1} dx$$

We will use a u -sub on the new integral. Let $u = x^2 + 1$, then $du = 2x dx$, so $\frac{1}{2} du = x dx$.

Our bounds become: $u(0) = (0)^2 + 1 = 1$ and $u(1) = (1)^2 + 1 = 2$

$$\begin{aligned} x \tan^{-1}(x) \Big|_0^1 - \int_0^1 \frac{x}{x^2 + 1} dx &= x \tan^{-1}(x) \Big|_0^1 - \frac{1}{2} \int_1^2 \frac{1}{u} du \\ &= x \tan^{-1}(x) \Big|_0^1 - \frac{1}{2} (\ln|u|) \Big|_1^2 = [1 \tan^{-1}(1) - 0 \tan^{-1}(0)] - \frac{1}{2} [\ln(2) - \ln(1)] \\ &= \left[\frac{\pi}{4} - 0 \right] - \frac{1}{2} [\ln(2) - 0] = \frac{\pi}{4} - \frac{1}{2} \ln(2) = \frac{\pi}{4} - \ln(\sqrt{2}) \end{aligned}$$

9. Find $\int \cos(\ln(x)) dx$

Let $u = \cos(\ln(x))$ and $dv = dx$.

These two equations imply $du = -\sin(\ln(x)) \frac{1}{x} dx$ & $v = x$.

Therefore, by IBP, we get

$$\int \cos(\ln(x)) dx = x \cos(\ln(x)) + \int \sin(\ln(x)) dx$$

On the new integral we let $u = \sin(\ln(x))$ and $dv = dx$

These two equations imply $du = \cos(\ln(x)) \frac{1}{x} dx$ & $v = x$.

Therefore, by IBP, we get

$$\begin{aligned} \int \cos(\ln(x)) dx &= x \cos(\ln(x)) + \int \sin(\ln(x)) dx \\ &= x \cos(\ln(x)) + \left[x \sin(\ln(x)) - \int \cos(\ln(x)) dx \right] \\ &= x \cos(\ln(x)) + x \sin(\ln(x)) - \int \cos(\ln(x)) dx \end{aligned}$$

We then note that the integral on the RHS is the same as the one we started with. Therefore,

$$2 \int \cos(\ln(x)) dx = x \cos(\ln(x)) + x \sin(\ln(x))$$

$$\int \cos(\ln(x)) dx = \frac{1}{2} [x \cos(\ln(x)) + x \sin(\ln(x))] + C$$

10. $\int 4x^4 \sin(2x) dx$

We can use Tabular IBP here since one of the factors in our integral is a polynomial and the other we can easily find antiderivatives for.

$P(x)$	$F(x)$	Sign +/-
$4x^4$	$\sin(2x)$	Skip
$16x^3$	$-\frac{1}{2}\cos(2x)$	+
$48x^2$	$-\frac{1}{4}\sin(2x)$	-
$96x$	$\frac{1}{8}\cos(2x)$	+
96	$\frac{1}{16}\sin(2x)$	-
0	$-\frac{1}{32}\cos(2x)$	+

Therefore, by Tabular IBP, we get

$$\int 4x^4 \sin(2x) dx = -2x^4 \cos(2x) + 4x^3 \sin(2x) + 6x^2 \cos(2x) - 6x \sin(2x) - 3 \cos(2x) + C$$

11. Find $\int x \tan^2(x) dx$

We let $u = x$ and $dv = \tan^2(x) dx$. Now, we have: $u = x \Leftrightarrow du = dx$ and

$dv = \tan^2(x) dx \Leftrightarrow dv = (\sec^2(x) - 1) dx \Leftrightarrow v = \tan(x) - x$. Now, we use the integration by parts to

finish the problem:

$$\int x \tan^2(x) dx = x(\tan(x) - x) - \int (\tan(x) - x) dx$$

$$= x \tan(x) - x^2 - \ln|\sec(x)| + \frac{1}{2}x^2 + C$$

$$= x \tan(x) - \ln|\sec(x)| - \frac{1}{2}x^2 + C.$$

12. Calculate the exact value of: $\int_{-1}^0 x \cdot 2^x dx$.

We can use Tabular IBP here since one of the factors in our integral is a polynomial and the other we can easily find antiderivatives for.

$P(x)$	$F(x)$	Sign +/-
x	2^x	Skip
1	$\frac{2^x}{\ln(2)}$	+
0	$\frac{2^x}{\ln(2)^2}$	-

Therefore, by Tabular IBP, we get

$$\int_{-1}^0 x \cdot 2^x dx = \left(\frac{x 2^x}{\ln(2)} - \frac{2^x}{[\ln(2)]^2} \right) \Big|_{-1}^0 = \left(-\frac{1}{[\ln(2)]^2} \right) - \left(\frac{-1}{2\ln(2)} - \frac{1}{2[\ln(2)]^2} \right) = \frac{1}{2\ln(2)} - \frac{1}{2[\ln(2)]^2}$$

13. Find $\int e^x \sin(2x) dx$.

We let $u = \sin(2x)$ and $dv = e^x dx$.

These two equations imply: $du = 2\cos(2x)dx$ and $v = e^x$.

Now, we apply the integration by parts formula and we obtain:

$$\int e^x \sin(2x) dx = \sin(2x)e^x - 2 \int e^x \cos(2x) dx.$$

Now, we turn our attention to $\int e^x \cos(2x) dx$.

We apply integration by parts on this integral. In particular we let $u = \cos(2x)$ and $dv = e^x dx$.

These two equations imply: $du = -2\sin(2x)dx$ and $v = e^x$.

Now, we apply the integration by parts formula and we obtain:

$$\int e^x \cos(2x) dx = \cos(2x)e^x + 2 \int e^x \sin(2x) dx.$$

Now, we know $\int e^x \sin(2x) dx = \sin(2x)e^x - 2 \int e^x \cos(2x) dx$. If we plug in what we just found for

$\int e^x \cos(2x) dx$ and solve for the desired integral we obtain:

$$\int e^x \sin(2x) dx = \sin(2x)e^x - 2 \left[\cos(2x)e^x + 2 \int e^x \sin(2x) dx \right]$$

$$\Leftrightarrow \int e^x \sin(2x) dx = \sin(2x)e^x - 2\cos(2x)e^x - 4 \int e^x \sin(2x) dx$$

$$\Leftrightarrow 5 \int e^x \sin(2x) dx = \sin(2x)e^x - 2\cos(2x)e^x$$

$$\Leftrightarrow \int e^x \sin(2x) dx = \frac{1}{5} [\sin(2x)e^x - 2\cos(2x)e^x] + C.$$